
Disentangled Representation Learning of Correlated Audio Attributes by Symmetry-Constrained Latent Subspaces

Mario Tan Yu
Center for Data Science
New York University
mty236@nyu.edu

Ishan Gupta
Center for Data Science
New York University
ig2671@nyu.edu

Adarsh Tiwari
Center for Data Science
New York University
adarsh.tiwari@nyu.edu

1 Problem Statement

Controllable audio representation learning is fundamentally limited by a conflict between probabilistic assumptions and physical reality. Most generative frameworks (VAE, GAN, Diffusion) rely on the assumption that latent factors are independent and identically distributed (i.i.d.). However, in musical signals, "style" (timbre) and "content" (pitch/rhythmic timing) are physically coupled through the resonant properties of the instrument. When a model is forced to map these correlated attributes into a standard i.i.d. latent space, it suffers from Semantic Leakage: a phenomenon where manipulating timbral identity inadvertently distorts the signal's pitch or micro-timing. This leakage indicates that current models learn surface-level statistical correlations rather than the underlying physical symmetries of the sound. **This project proposes that the "Inconsistency of Intended Semantics" can only be resolved by abandoning pure unsupervised i.i.d. priors in favor of a Self-Supervised Causal Framework. By explicitly enforcing group-theory constraints—specifically pitch-shift equivariance—we aim to bake the physical laws of audio into the latent manifold.** We investigate whether this symmetry-constrained approach can mitigate semantic leakage and provide the independent control required for high-fidelity audio manipulation.

2 Dataset and Data Statistics

We use a mix of two different audio datasets to help the model tell the difference between the notes being played and the unique sound of the bass itself. We focus on 'Source Identity' - treating each unique recording as its own style. This gives the model thousands of different examples to learn from while still giving us a clear way to check if it is getting the notes right (providing the model with a denser distribution of latent space while maintaining clear ground truth for evaluation later). Unlike vocals, where phonemes (speech sounds) constantly alter the spectral envelope, we will focus on bass guitar because it maintains a consistent resonant "fingerprint" across its range. This allows the model to more easily separate the periodic components (Pitch) from the fixed filter characteristics of the instrument's body and amplifier (Style). Hence our dual-dataset approach to isolate the latent factors of pitch and timbre:

- **NSynth (Bass Subset):** We utilize 68,955 monophonic (isolated bass) notes. Each sample includes a MIDI pitch label, providing the ground truth required to calculate the Mutual Information Gap (MIG) and ensure our pitch vector z_c does not "leak" into the style latent z_s .
- **MoisesDB (2025):** We utilize 240 professional bass stems. These represent 45 unique "Source Identities." We use these tracks to train for **Identity Invariance** via a self-supervised pitch-shifting loop, forcing the style encoder to remain stationary across performance-level variations (e.g., slides and vibrato).

35 • **Zero-Shot Evaluation Set:** To test generalizability, we reserve a small set of isolated
 36 bass tracks from famous recordings (e.g., Pink Floyd’s *Money*). These are used as out-of-
 37 distribution probes to verify if the style subspace z_s remains stationary across varied melodic
 38 phrases.

39 3 Methodology

40 We propose a *Symmetry-Constrained VAE* that partitions the latent space into style (z_s) and content
 41 (z_c) components by enforcing group-theoretic priors. Unlike standard generative models, we explicitly
 42 bake the physics of pitch-shifting into the loss function. Our model defines a pitch-shift group action
 43 G where an input transformation T_g (shifting frequency) must map to specific latent behaviors:

- 44 • **Style Invariance:** $z_s(T_g x) = z_s(x)$. The instrument identity must remain stationary under
 45 pitch transformations.
- 46 • **Content Equivariance:** $z_c(T_g x) = \rho(g)z_c(x)$. The melodic representation must move
 47 linearly with the shift g .

We optimize a composite loss function to minimize “Semantic Leakage”:

$$\mathcal{L} = \mathcal{L}_{recon} + \beta \text{KL}(q(z|x)||p(z)) + \lambda \mathcal{L}_{sym}$$

48 where $\mathcal{L}_{sym} = \|z_s(x) - z_s(T_g x)\|^2$ is the symmetry penalty. We evaluate this approach against two
 49 primary baselines:

- 50 1. **β -VAE:** A standard disentanglement baseline that relies on a high KL-penalty ($\beta > 1$) to
 51 force statistical independence without physical priors.
- 52 2. **Autoregressive Hierarchical VAE:** A baseline that models local temporal dependencies
 53 but often entangles style across different levels of the hierarchy.

54 4 Proposed Metrics

55 We evaluate disentanglement through three quantitative frameworks, using **MIDI Pitch** (from NSynth)
 56 and **Track ID** (from MoisesDB) as our ground-truth factors (v_k).

- 57 • **Mutual Information Gap (MIG):** We calculate the normalized difference between the
 58 two highest mutual information values between a factor v_k and all latent dimensions z_i . A
 59 high MIG score confirms that pitch and style are captured by unique, non-overlapping latent
 60 coordinates.
- 61 • **DCI Framework:** We report **Disentanglement** (dimension purity), **Completeness** (factor
 62 concentration), and **Informativeness** (prediction accuracy). While β -VAE often achieves
 63 high Disentanglement at the cost of Informativeness, we hypothesize our symmetry con-
 64 straint will maintain high factor-prediction accuracy without "leaking" pitch into the style
 65 latent.
- 66 • **Signal-to-Reconstruction Ratio (SRR):** To ensure that enforcing symmetry does not
 67 degrade audio quality, we measure the log-spectral distance between the original and
 68 reconstructed spectrograms.

69 We anticipate that our symmetry-constrained model will outperform baselines in MIG and DCI-D
 70 scores, particularly on out-of-distribution "Famous" test clips where standard VAEs typically suffer
 71 from semantic leakage.

Table 1: Projected Evaluation Metrics across Baselines and Proposed Model

Model	MIG ($\uparrow$)	DCI-D ($\uparrow$)	Informativeness ($\uparrow$)	SRR (dB) ($\uparrow$)
Standard β -VAE	0.12	0.34	0.65	18.2
Hierarchical VAE	0.22	0.48	0.72	16.5
Ours (Symmetry-Constrained)	0.45	0.78	0.84	15.8

5 Related Work

References

- [1] Gino Brunner et al. Midi-vae: Modeling dynamics and particle tracks in music. In *International Society for Music Information Retrieval (ISMIR)*, 2018.
- [2] Ricky TQ Chen, Xuechen Li, Roger B Grosse, and David K Duvenaud. Isolating sources of disentanglement in vaes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [3] Jade Copet et al. Simple and controllable music generation. *arXiv preprint arXiv:2306.05284*, 2023.
- [4] Prafulla Dhariwal et al. Jukebox: A generative model for music. *arXiv preprint arXiv:2005.00341*, 2020.
- [5] Cian Eastwood and Christopher KI Williams. A framework for the quantitative evaluation of disentangled representations. In *International Conference on Learning Representations (ICLR)*, 2018.
- [6] Jesse Engel et al. Ddsp: Differentiable digital signal processing. In *International Conference on Learning Representations (ICLR)*, 2020.
- [7] Jesse Engel, Cinjon Resnick, Adam Roberts, Sander Dieleman, Mohammad Norouzi, Douglas Eck, and Karen Simonyan. Neural audio synthesis of musical notes with wavenet autoencoders. In *International Conference on Machine Learning (ICML)*, 2017.
- [8] Ian Goodfellow et al. Generative adversarial nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
- [9] Irina Higgins, David Amos, David Pfau, Terrence Okada, Loic Matthey, Demis Hassabis, and Alexander Lerchner. Towards a definition of disentangled representations. *arXiv preprint arXiv:1812.02230*, 2018.
- [10] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [11] Cheng-Zhi Anna Huang, Ashish Vaswani, Jakob Uszkoreit, Noam Shazeer, Curtis Hawthorne, Andrew M Dai, Matthew D Hoffman, and Douglas Eck. Music transformer: Generating music with long-term structure. *arXiv preprint arXiv:1809.04281*, 2018.
- [12] Jaesung Jeon et al. Mge-ldm: Joint latent diffusion for simultaneous music generation and source extraction. *arXiv preprint arXiv:2403.14151*, 2024.
- [13] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations (ICLR)*, 2014.
- [14] Francesco Locatello et al. Challenging common assumptions in the unsupervised learning of disentangled representations. In *International Conference on Machine Learning (ICML)*, 2019.
- [15] Yusong Luo et al. Dismix: Disentangling mixtures of musical instruments for source-level pitch and timbre manipulation. In *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.

- 110 [16] Meinard Müller. *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications*.
111 Springer, 2021.
- 112 [17] Wei-An Pang et al. Disentangled representation learning. *Tsinghua University Press*, 2020.
- 113 [18] Ashis Pati and Alexander Lerch. Is disentanglement enough? on latent representations for
114 controllable music generation. In *Proceedings of the 22nd International Society for Music*
115 *Information Retrieval Conference (ISMIR)*, pages 513–520, 2021.
- 116 [19] Adam Roberts, Jesse Engel, Colin Raffel, Curtis Hawthorne, and Douglas Eck. A hierarchical
117 latent vector model for learning long-term structure in music. In *International Conference on*
118 *Machine Learning (ICML)*, 2018.
- 119 [20] Prem Seetharaman et al. Evaluation of latent space disentanglement in the presence of in-
120 terdependent attributes. In *International Society for Music Information Retrieval (ISMIR)*,
121 2020.
- 122 [21] Zachary Wang et al. Ditto-2: Distilled diffusion inference-time t-optimization for music
123 generation. *arXiv preprint*, 2024.